Asymptotic Analysis Comprehensions

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Contents

Figure 1: From order evaluating to asymptopia ¡—

1 Basics of order, $\mathcal{O}$ and o

1.1 Summary/Comprehensions

The knowledge of orders of 2 infinitesimal or infinite quantities and the connection between them are common for us. Here we list some notes you may need to pay attention to:

1. Sometimes the "big $\mathcal{O}$ constant" in a strong function can be related to the argument, and the same is true for the small "o".

Example. $\sin(xy) = O(1)$, $\sin(xy) = O_y(x)$

2. The infinitesimal of 2 functions can't always keep for their inverse functions, i.e. $\varphi(x) = o(f(x)) \nRightarrow \widetilde{f}(x) = o(\widetilde{\varphi}(x))$. (Here: $\varphi(x)$ and $f(x)$ are increasing functions, as $\widetilde{\varphi}(x)$ and $\widetilde{f}(x)$ are their inverse functions)

Example. $f(x) = e^x, \quad \varphi(x) = \frac{e^x}{x}, \quad x \to \infty$

3. The elementary of $\mathcal{O}$ and o notation:

Notes. $\varphi = O(f)$ means an inequality, while $\varphi = o(g)$ express a limitation.

4. 16 Rules of calculating $\mathcal{O}$ and o:

1. Transitivity

   [1] $\lim_{x\to x_0} f(x) = \infty$, $\varphi(x) = O(1) \implies \varphi(x) = o(f(x))$

   [2] $f = O(\varphi)$, $\varphi = O(\psi) \implies f = O(\psi)$

   [3] $f = O(\varphi)$, $\varphi = o(\psi) \implies f = o(\psi)$

2. $\mathcal{O}$

   [4] $O(f) + O(g) = O(f+g)$

   [5] $O(f)O(g) = O(fg)$

3. $\mathcal{O}$ & o

   [6] $o(1)O(f) = o(f)$

   [7] $O(1)o(f) = o(f)$

   [8] $O(f) + o(f) = O(f)$

4. o

   [9] $o(f) + o(g) = o(|f| + |g|)$

   [10] $o(f)o(g) = o(fg)$

5. Power

   [11] $(O(f))^k = O_k(f^k),\ k \in \mathbb{N}$

   [12] $(o(f))^k = o(f^k)$

6. Equivalence $\sim$

   [13] $f \sim g$, $g \sim \varphi \implies f \sim \varphi$

6. Equivalence $\sim$

   [14] $f = o(g),\ g \sim \varphi \implies g \sim \varphi \pm kf$

7. Summation of o

   [15] $f,\ g > 0,\ f = o(g),\ x \to \infty \implies \int_A^B f(x)dx = o(\int_A^B g(x)dx)$, $B > A \to \infty$
   Specially, if $\exists A_0$ s.t. $\int_{A_0}^{\infty} g dx < \infty$, then $\int_A^{\infty} f(x)dx = o(\int_A^{\infty} g(x)dx)$, $B > A \to \infty$

   [16] $a_n,\ b_n > 0,\ a_n = o(b_n),\ n \to \infty \implies \sum_{n=N}^{M} a_n = o(\sum_{n=N}^{M} b_n)$, $M > N \to \infty$
   Specially, if $\sum_{n=1}^{\infty} b_n < \infty$, then $\sum_{n=N}^{\infty} a_n = o(\sum_{n=N}^{\infty} b_n)$, $N \to \infty$

5. Extension for Taylor's formula: /Tech/

Theorem 1.1. *If $f^{(n)}(x)$ exists on $U(x_0;\delta)$ and $|f^{(n)}(x)| \le M$, then*

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + O(|(x - x_0)^n|)$$

holds on $U(x_0;\delta)$.

Corollary 1.1. *1.* $\sin x = x - \frac{x^3}{3!} + O(x^5)$, $x \to 0$ *or* $|x| < \infty$

2. $\cos x = 1 - \frac{x^2}{2!} + O(x^4)$, $x \to 0$ *or* $|x| < \infty$

3. $\log(1 + x) = x - \frac{x^2}{2} + O(x^3)$, $x \to 0$

4. $(1 + x)^{\alpha} = 1 + \alpha x + O(x^2)$, $x \to 0$

5. $e^x = 1 + x + O(x^2)$, $x \to 0$ *or* $|x| < 1$

Notes. 1. Often $\log(1 + x) = x + O(x^2)$ is enough to use.
2. $\tan x = 1 + \frac{x^3}{3} + O(x^5)$.

Corollary 1.2. *If f satisfies $\lim_{x \to x_0} f(x) = 0$, then*

1. $\sin f(x) = f(x) - \frac{f(x)^3}{3!} + O(f(x)^5)$, $f(x) \to 0$ *or* $|f(x)| < \infty$

2. $\cos f(x) = 1 - \frac{f(x)^2}{2!} + O(f(x)^4)$, $f(x) \to 0$ *or* $|f(x)| < \infty$

3. $\log(1 + f(x)) = f(x) - \frac{f(x)^2}{2} + O(f(x)^3)$, $f(x) \to 0$

4. $(1 + f(x))^{\alpha} = 1 + \alpha f(x) + O(f(x)^2)$, $f(x) \to 0$

5. $e^f(x) = 1 + f(x) + O(f(x)^2)$, $f(x) \to 0$ *or* $|f(x)| < 1$

1.2 Problem Sets

Problem. (Infinitesimal calculation) $\int_{x^{-2}}^{\infty} e^{-t}\log t\,dt = o(1),\ x \to 0.$

> *Solution.* $\int_{x^{-2}}^{\infty} e^{-t}\log t\,dt < \int_{x^{-2}}^{\infty} e^{-\frac{t}{2}}\,dt = -2e^{-\frac{t}{2}}|_{x^{-2}}^{\infty} = 2e^{-\frac{1}{2x^2}}$

Problem.

$$(\text{Classical limit review})x^A = o((1+a)^{\epsilon x}),\ x \to \infty,\ \text{as } \epsilon, a > 0\ \forall A.$$
$$(\text{Develop})(\log(x))^A = o(x^{\epsilon}),\ x \to \infty,$$
$$(f(x))^A = o(e^{\epsilon f(x)}),\ x \to \infty.$$

> *Solution.* Let $n \triangleq [x],\ m \triangleq [A]+1$, then $(1+a)^x \geq (1+a)^n \geq C_n^{m+1}a^{m+1}(x \to \infty$, i.e. $n \to \infty$, and $n > 2m+1) \geq \frac{(n-m)^{m+1}}{(m+1)!}a^{m+1} \geq \frac{n^{m+1}}{2^{m+1}(m+1)!}a^{m+1}$, then $\frac{(1+a)^x}{x^A} \geq \frac{n^{m+1}a^{m+1}}{2^{m+1}(m+1)!(n+1)^m} \geq \frac{na^{m+1}}{2^{2m+1}(m+1)!}$, then $\lim_{x\to\infty}\frac{x^A}{(1+a)^x} = 0$, *i.e.* $x^A = o((1+a)^x),\ x \to \infty$. Since $\lim_{x\to\infty}\frac{x^A}{(1+a)^{\epsilon x}} = \lim_{x\to\infty}(\frac{x^{\frac{A}{\epsilon}}}{(1+a)^x})^{\epsilon} = 0$, then $x^A = o((1+a)^{\epsilon x}),\ x \to \infty$.

Problem. /Tech tool/

Let $a_n = O(b_n),\ n \geq 1$, and series $\sum_{n=1}^{\infty} b_n < \infty$, then $\exists C$, s.t. $\sum_{k=1}^{n} a_k = C + O\left(\sum_{k=n+1}^{\infty} b_k\right)$.

> *Solution.* Obviously, series $\sum_{k=1}^{\infty} a_k$ converges, then $\exists C$ s.t. $\sum_{k=1}^{\infty} a_k = C$, then $\sum_{k=1}^{n} a_k = \sum_{k=1}^{\infty} a_k - \sum_{k=n+1}^{\infty} a_k = C + O\left(\sum_{k=n+1}^{\infty} b_k\right)$.

Proposition. /Tech tip/

The 1st principle of order evaluating (specially need using Taylor's series to evaluate) is paying more attension on and catching the infinitesimal or infinite quantities which those are not so good to distinguish directly "by eyes".

Here are some useful techs and interesting examples:

1. /Tech tool/ $\sqrt[n]{n} = e^{\frac{\log n}{n}} = 1 + \frac{\log n}{n} + O(\frac{\log^2 n}{n^2}),\ n \to \infty$
2. Using 1, you could get $(n+a)^{n+b} = n^{n+b}e^a(1+\frac{a(b-\frac{a}{2})}{n}+O(\frac{1}{n^2})) \Longrightarrow (1+\frac{a}{n})^{n+b} = e^a(1+\frac{a(b-\frac{a}{2})}{n}+O(\frac{1}{n^2}))$ (development of $\lim_{n\to\infty}(1+\frac{1}{n})^n = e$).
3. Layer-by-layer

$$\begin{aligned}(x+(x+(x^{\frac{\alpha}{2}}))^{\frac{1}{2}})^{\frac{1}{2}} &= x^{\frac{1}{2}}(1+(x^{-1}+x^{\frac{\alpha}{2}-2})^{\frac{1}{2}})^{\frac{1}{2}}\ (0<\alpha<2)\\ &= x^{\frac{1}{2}}(1+\frac{1}{2}(x^{-1}+x^{\frac{\alpha}{2}-2})^{\frac{1}{2}}+O(x^{-1}))\\ &= x^{\frac{1}{2}}(1+\frac{1}{2}x^{-\frac{1}{2}}(1+x^{\frac{\alpha}{2}-1})^{\frac{1}{2}}+O(x^{-1}))\\ &= x^{\frac{1}{2}}+\frac{1}{2}(1+\frac{1}{2}x^{\frac{\alpha}{2}-1}+O(x^{\alpha-2}))+O(x^{-1})\end{aligned}$$

4. $\lim_{n\to\infty} \cos^n \frac{x}{\sqrt{n}} = \lim_{n\to\infty} e^{n\log(1-\frac{x^2}{2n}+O_x(\frac{1}{n^2}))} = e^{-\frac{x^2}{2}}(1+O_x(\frac{1}{n}))$

   Related (just form): Wallis' integral $W_n = \int_0^{\frac{\pi}{2}} (\cos x)^n dx = \int_0^{\frac{\pi}{2}} (\sin x)^n dx = \frac{n-1}{n} W_{n-2} = \frac{(n-1)!!}{n!!} (\frac{\pi}{2})^{\delta_{(n(mod2))0}} = \sqrt{\frac{\pi}{2n}}(1+o(1))$, $(n \to \infty)$

2 Appendix

2.1 Extending tests for convergence of number series

2.1.1 Talking from Cauchy Condensation Test

Theorem 2.1. *(Caucy Condensation Test) Tech toolkit*
$\{a_n\}_n^\infty \searrow^+$ *(i.e.* $\cdots \geq a_n \geq a_{n-1} \geq \cdots \geq a_1 \geq 0$*), then* $\sum_{n=1}^\infty a_n$ *and* $\sum_{k=1}^\infty 2^k a_{2^k}$ *have same convergence.*

Connections betwwen Condensation test and Integral test:

$$\sum_{n=1}^{\infty} a_n < \infty \iff \int_1^\infty f(x)dx = \infty \text{ substitute with: } x = 2^y$$
$$\iff \int_1^\infty 2^y f(2^y)dy < \infty \iff \sum_{k=1}^{\infty} 2^k a_{2^k} < \infty$$

Theorem 2.2. *(Maclaurin-Cauchy Integral Test) Tech toolkit*
$f \searrow^+$ *on* $[1, +\infty)$*, then* $\sum_{n=1}^\infty f(n)$ *and* $\int_1^\infty f(x)dx$ *have same convergence.*

Here are some examples we could use Condensation test to solve:

1. $\sum_{n=1}^\infty \frac{1}{n^p}$, $\sum_{n=2}^\infty \frac{1}{n \log^p n}$, $\sum_{n=2}^\infty \frac{1}{(\log n)^p}$ (combine Cauchy root test), $\sum_{n=2}^\infty \frac{\log n}{n^2}$
2. $\sum_{n=16}^\infty \frac{1}{n(\log n)(\log\log n)(\log\log\log n)^p}$
3. More general, let's consider $f(x) := n^{-a}(\log n)^{-b}(\log\log n)^{-c}$, $\sum f(n) < \infty$, $a > 1$, when $a = 1$ condensation transformation gives $\sum n^{-b}(\log n)^{-c}$ (seems like the logarithms "shift to the left") so when $a = 1$, it converges as $b > 1$. When $b = 1$, the value of c enters.
   This result readily generalizes: the condensation test, applied repeatedly, can be used to show that for $k = 1, 2, 3, \cdots$, the generalized Bertrand series $\sum_{n \geq N} \frac{1}{n \cdot \log n \cdot \log\log n \cdots \log^{\circ(k-1)} n \cdot (\log^{\circ k} n)^\alpha}$ $(N = \lfloor \exp^{\circ k}(0) \rfloor + 1)$ converges for $\alpha > 1$ and diverges for $0 \leq \alpha \leq 1$.
   from Wikipedia: Cauchy Condensation Test

Besides, the Cauchy condensation test is a partial case of following classical result deu to Schlomilch:

Theorem 2.3. *(Schlomilch)*
$\{a_n\}_n^\infty \searrow^+$, $\{u_n\}_0^\infty : u_0 < u_1 < \cdots$ *and* $\frac{\Delta u_n}{\Delta u_{n-1}} \leq C$*, then* $\sum_{n=1}^\infty a_n$ *and* $\sum_{n=1}^\infty \Delta u_n a_{u_n} = \sum_{n=1}^\infty (u_{n+1} - u_n) a_{u_k}$ *have same convergence.*

Notes. 1. You may note it that let $u_n = 2^n$, it's Cauchy condensation test. Moreover, you could also use $u_n = p^n$, $p > 1$.

2.2 Recall history of tests with monotonicity assumption for sequence